

# REDUCING CUSTOMER CHURN THROUGH DATA-DRIVEN RETENTION

Advanced Analytics Engagement | Confidential



100,000 customers  
analyzed



103 variables



Gradient Boosting model  
(AUC 0.69)



Prepared for Executive Leadership

Nearly **50%** of customers churn annually —  
putting **\$35M** in revenue at risk

# Executive Summary

Situation • Complication • Resolution



## THE PROBLEM

At-risk users churn across **100,000** subscribers — equivalent to **\$35M** annual revenue leakage



## ROOT CAUSE

Equipment age (#1 driver), declining usage, and bill overages are the leading predictive signals — not demographics alone



## ANALYTICAL APPROACH

Merged behavioral + demographic data, engineered **103** features, benchmarked **4 ML** models — Gradient Boosting achieves **0.69 AUC**



## KEY OPPORTUNITY

High-risk risk segment (declining revenue + declining usage) shows **53%** churn vs **43%** in upsell segment



## RECOMMENDED ACTION

Targeted retention program can save **\$5M/yr** with **15%** improvement in at-risk retention



**\$35M**

Revenue at Risk



**50%**

Churn Rate



**0.69**

Model AUC



**\$5M**

15% Lift Upside



## Telecom retention is a \$35M problem – acquisition costs 5–7x retention spend



### INDUSTRY BENCHMARKS

- Wireless carriers lose 1.5–2.0% of subscribers monthly
- Customer acquisition cost: \$300–400 per subscriber
- Retention cost: \$50–75 per save — 5–7x ROI advantage
- Churned customers take 18+ months of margin to replace



### CLIENT SITUATION

- 100,000 active subscribers in analysis cohort
- 49.6% observed churn — near coin-flip attrition
- \$58 avg monthly rev per churned customer
- \$35M annual revenue directly at risk



### COMPLICATION

No systematic early-warning system



### GAP

Retention spend not data-targeted



### IMPERATIVE

Analytics-led intervention needed



Nearly **50%** of customers churn annually – putting **\$35M** in revenue at risk



100,000 customers analyzed

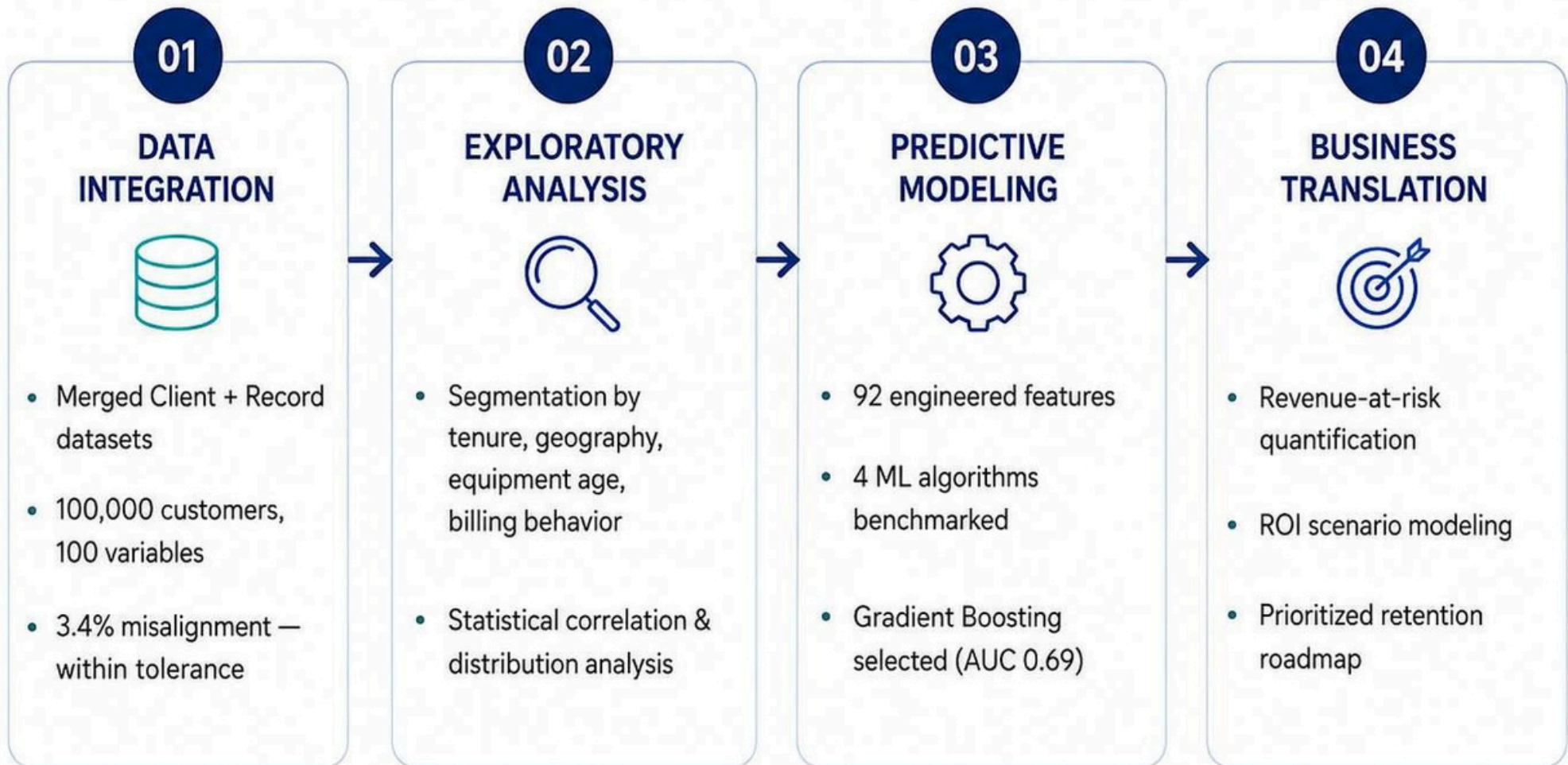


103 variables



Gradient Boosting model (AUC 0.69)

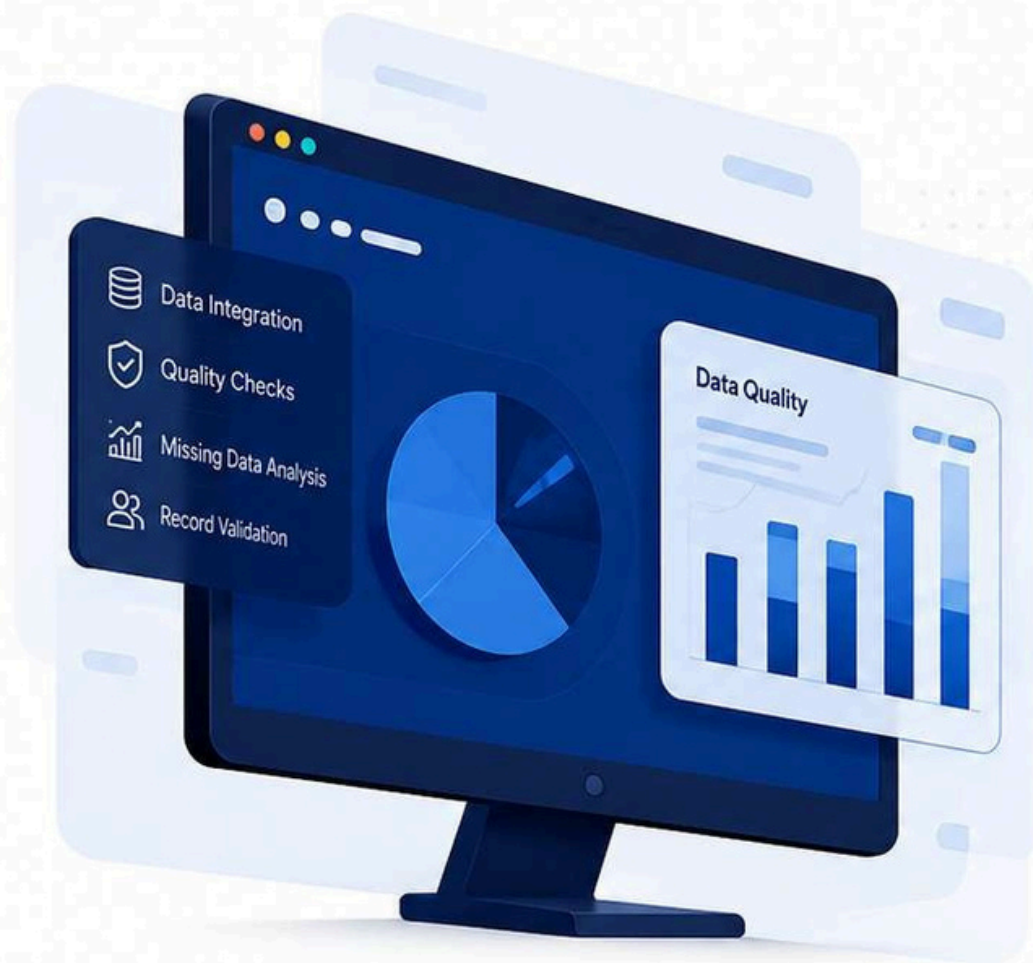
Our analytical approach follows a rigorous **four-phase** consulting framework





## Data Overview | Quality Assessment

**100,000** customer records provide a robust foundation for predictive analytics



| DATA QUALITY SCORECARD |         |
|------------------------|---------|
| Total Records          | 100,000 |
| Total Features         | 100     |
| Features Used (ML)     | 92      |
| Missing Data Rate      | 3.4%    |
| Columns w/ Missing     | 43      |
| Churn Rate             | 49.6%   |
| Churned                | 49,562  |
| Retained               | 50,438  |

# Business Impact | Financial Quantification

## \$35M in annual revenue is directly at risk from customer churn



### REVENUE AT RISK

50,000 churned customer generate **\$35M** in revenue for company — direct impact on bottom line from high-value accounts.



### RETENTION ROI

A 15% improvement saves **\$5M** annually.  
A 30% improvement could save **\$10M+** yearly, investments in targeted retention yield a back within 2–3 months.



### UPLIFT IMPACT

Industry best-in-class carriers achieve 35–50% annual uplift.  
Closing half the gap represents **\$18M+** opportunity.

### ANNUAL REVENUE AT RISK





## Key Finding #1 | Equipment Lifecycle

Handset age is the #1 churn predictor — customers with **24+ month** devices are highest risk

### IMPLICATION



Equipment age (eqpdcyd) ranks #1 in both correlation analysis and ML feature importance.



Customers holding devices 24+ months show significantly elevated churn — likely driven by technology obsolescence, battery degradation, and competitive upgrade efforts.



Launch subsidized upgrade program for devices **>18 months**



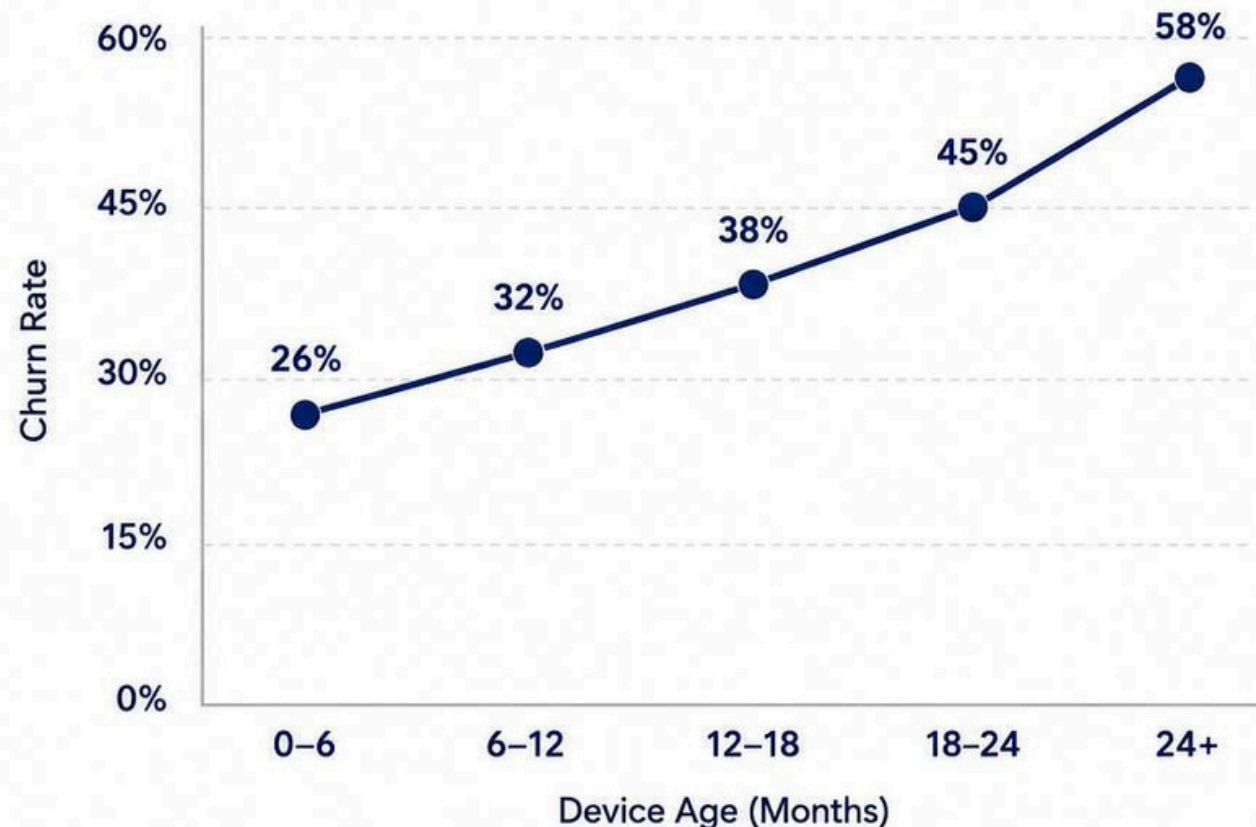
Trigger upgrade offer at **12-month** anniversary



**Bundle upgrade** with plan renewal for dual lock-in

★ Model Importance: eqpdys = 15.1% of predictive power

CHURN RATE BY DEVICE AGE (MONTHS)



# Key Finding #2 | Behavioral Churn Matrix

Declining revenue + declining usage =  
**53%** churn — the critical intervention segment

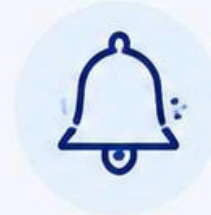




## Key Finding #3 | Bill Shock Analysis

**Bill overages are a leading churn indicator – proactive alerts can prevent attrition**

| KEY METRICS             |         |
|-------------------------|---------|
| Customers with overages | 57%     |
| C0%                     | 49.2%   |
| Churn (with overages)   | 49.8%   |
| Incremental risk        | +0.6 pp |



Deploy real-time overage alerts at **80%** plan threshold



Offer automatic plan upgrade before bill cycle closes

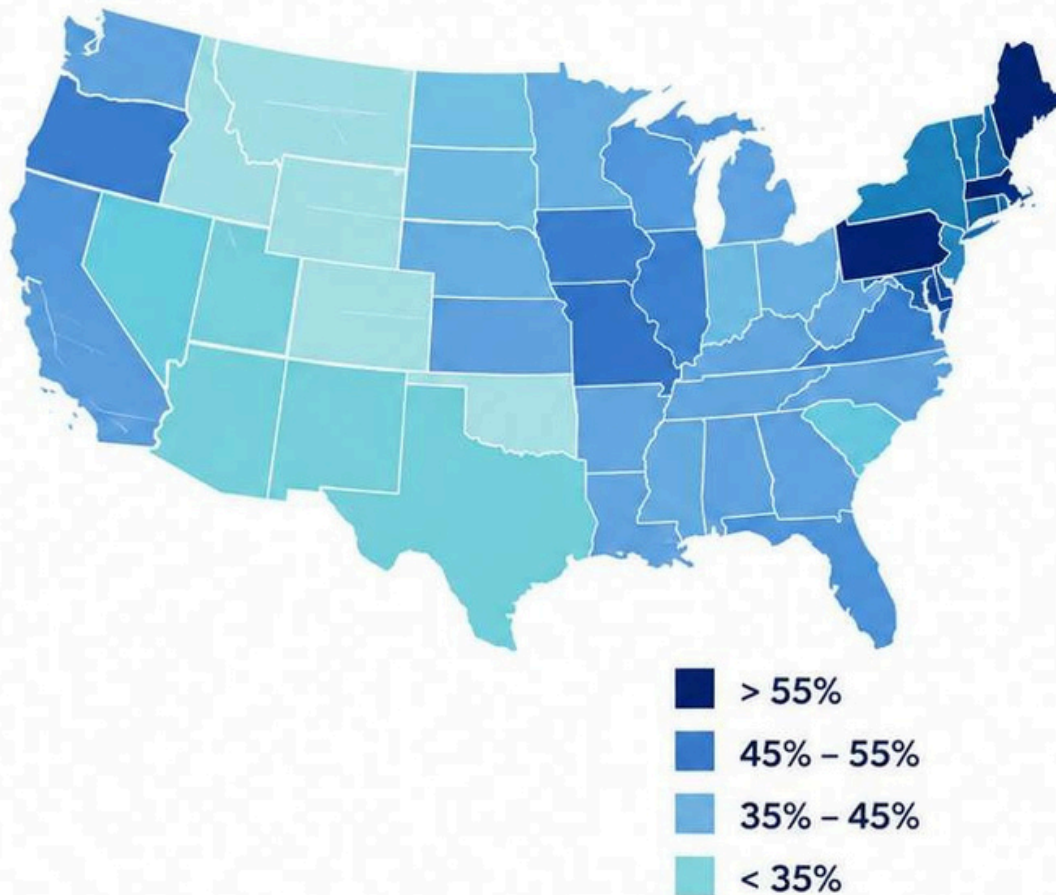


Cap overage charges for loyalty-tier customers

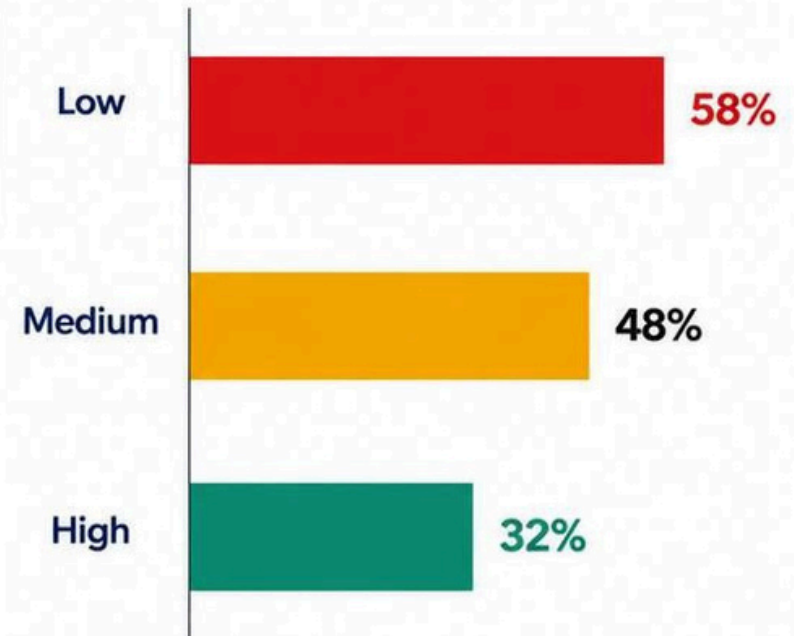
## Segmentation | Where to Focus

Churn varies significantly by geography  
and credit class — retention must be localized

CHURN RATE BY REGION



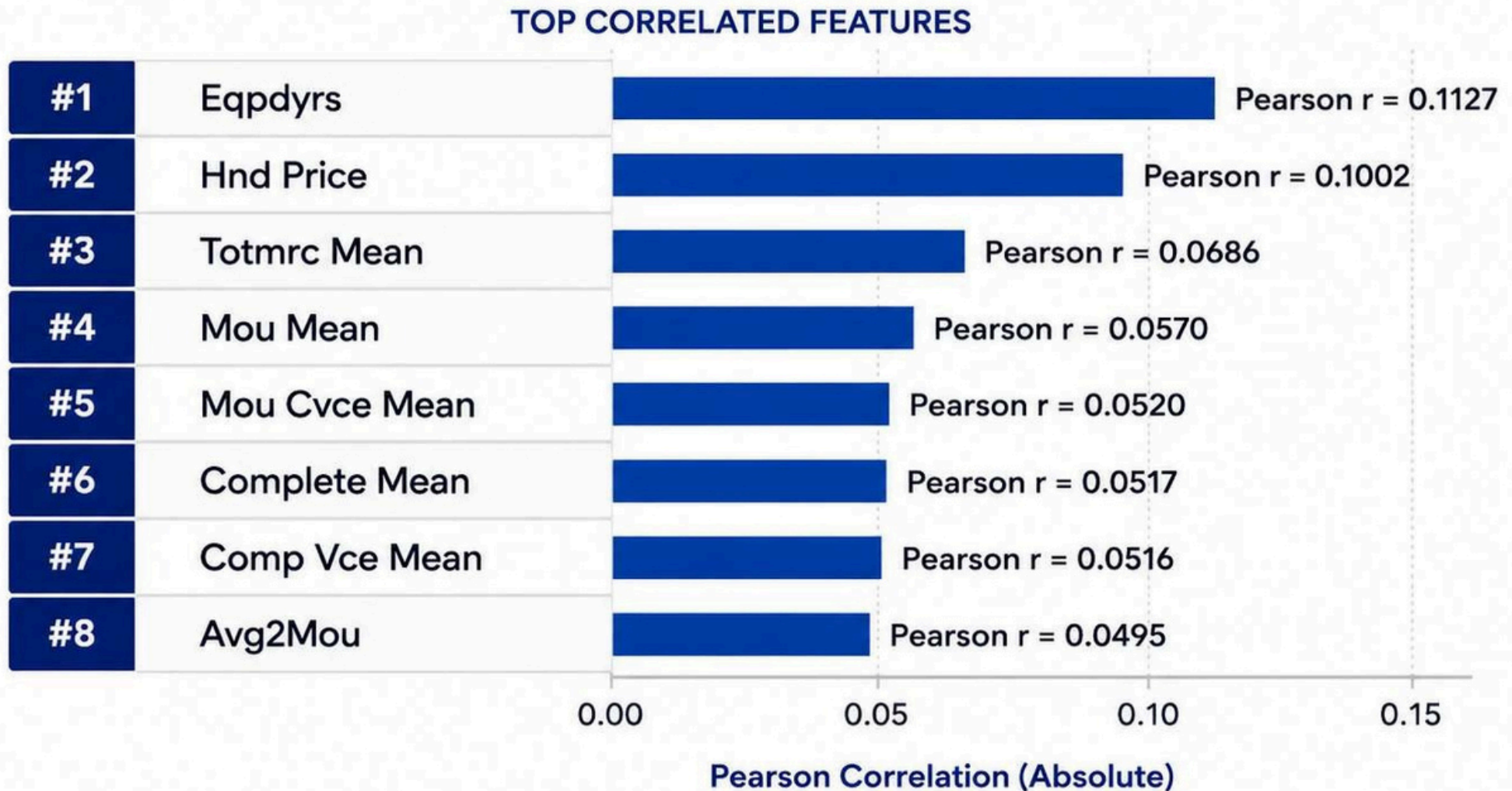
CHURN RATE BY CREDIT CLASS





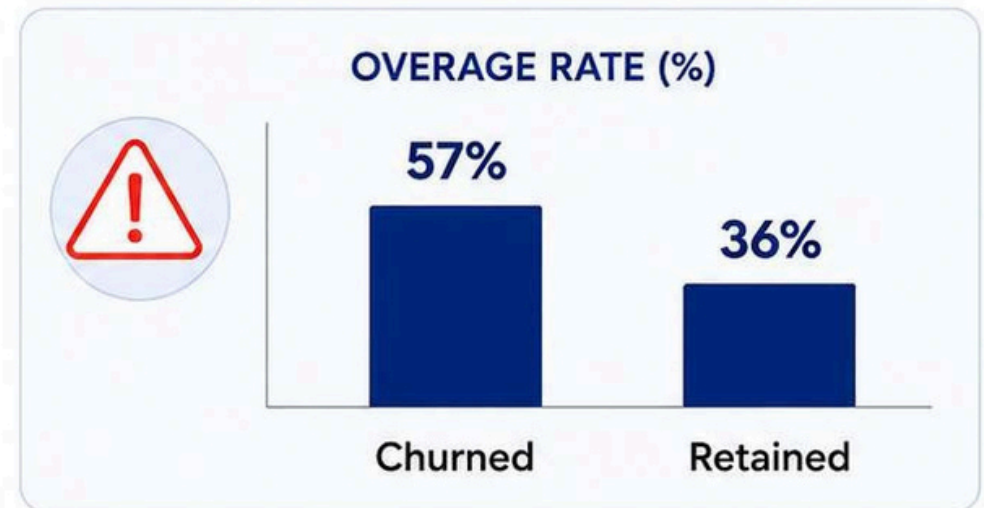
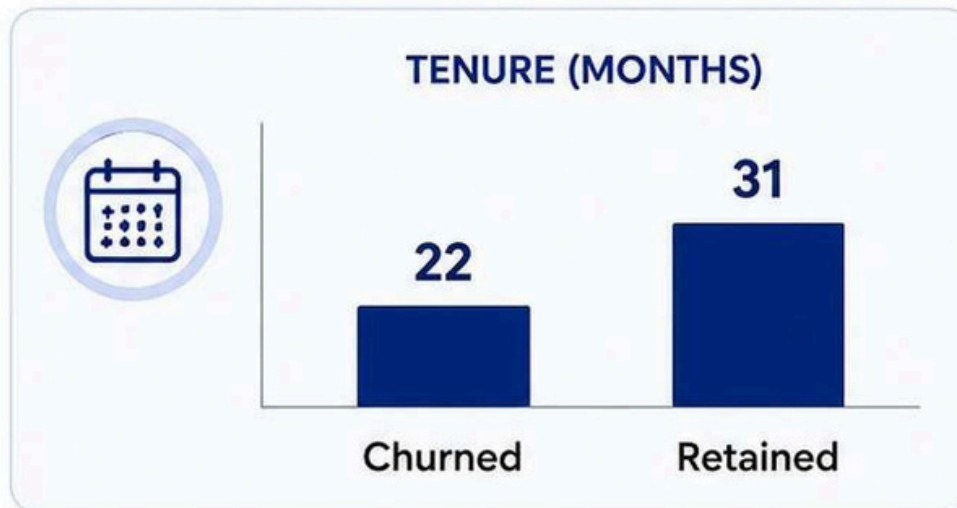
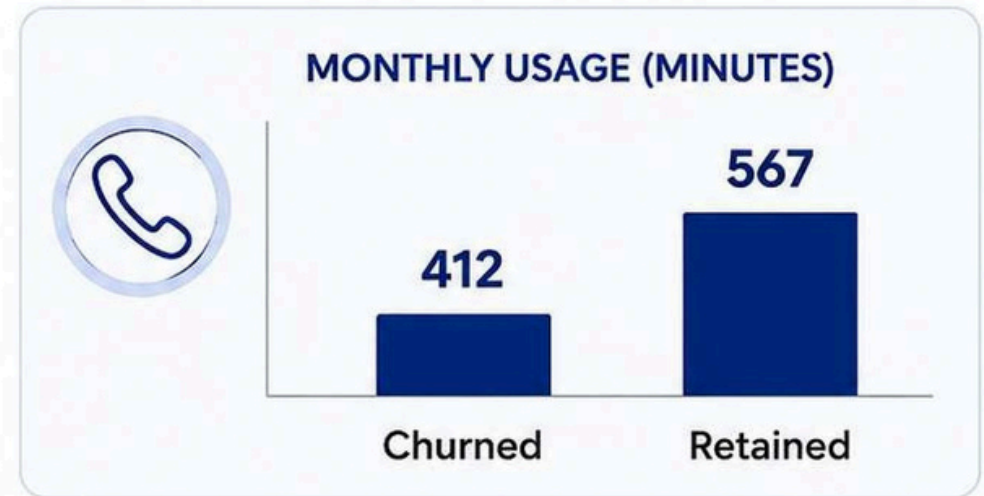
## Feature Analysis | Correlation with Churn

Statistical analysis confirms equipment age, tenure, and usage as top churn drivers



## Behavioral Profiles | Churned vs Retained

Churned customers show distinct behavioral profiles across usage, revenue, and tenure





# Predictive Modeling | Model Benchmarking

Gradient Boosting delivers 0.69 AUC — sufficient for effective at-risk customer targeting

| Model               | AUC    | F1     | Precision | Recall | Accuracy |
|---------------------|--------|--------|-----------|--------|----------|
| Logistic Regression | 0.6307 | 0.5802 | 0.5842    | 0.5764 | 0.5967   |
| Random Forest       | 0.6687 | 0.6283 | 0.6035    | 0.6552 | 0.6158   |
| ★ Gradient Boosting | 0.6915 | 0.6367 | 0.6288    | 0.6449 | 0.6353   |
| XGBoost             | 0.6713 | 0.6223 | 0.6161    | 0.6366 | 0.6230   |



0.69 AUC

Good Discrimination



0.64 Recall

Identifies At-Risk Customers

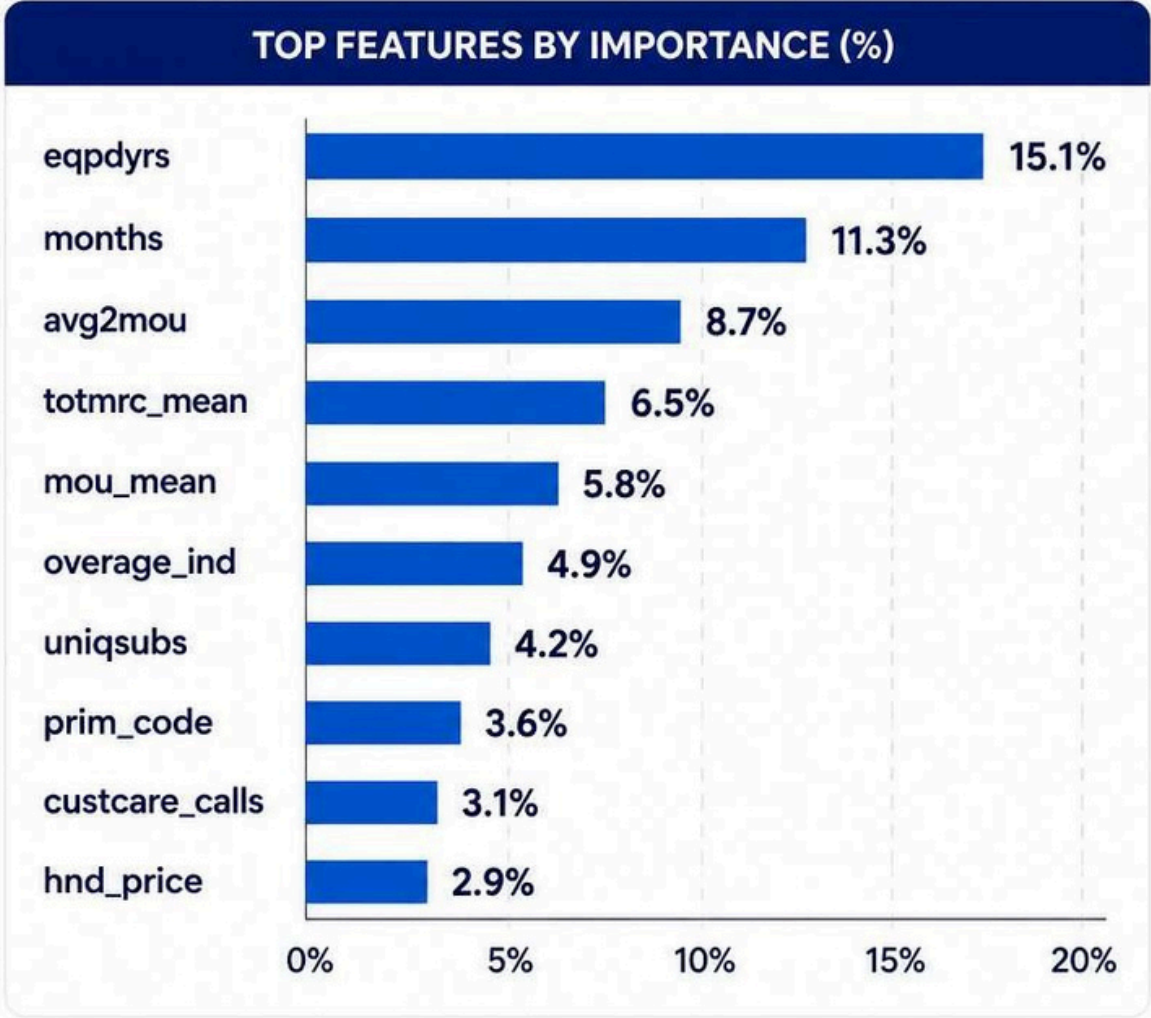


Balanced Performance

Across All Metrics

# Feature Importance | What Drives Predictions

Equipment age, tenure, and usage trends dominate the predictive model



| BUSINESS INTERPRETATION                                                               |             |                                                        |
|---------------------------------------------------------------------------------------|-------------|--------------------------------------------------------|
|    | eqpdyrs     | Handset age — older devices = higher churn probability |
|    | months      | Tenure — newer customers are more likely to churn      |
|    | avg2mou     | Usage decline — reduced engagement increases risk      |
|  | overage_ind | Bill shock — financial stress drives churn             |
|  | totmrc_mean | Lower revenue — less committed customers               |



# Implementation Roadmap & Next Steps

A phased approach to deliver measurable impact.



## PHASE 1

0–3 Months

### BUILD & DEPLOY

- Deploy churn scoring engine
- Integrate data pipelines
- Create risk dashboards



## PHASE 2

3–6 Months

### INTERVENE & ENGAGE

- Launch targeted retention campaigns
- Device upgrade offers
- Bill shock alerts & plan optimization
- Proactive outreach to high-risk customers



## PHASE 3

6–12 Months

### MEASURE & OPTIMIZE

- Track KPIs & campaign performance
- A/B test interventions
- Refine models & scale successful strategies

### EXPECTED OUTCOME



## 5% CHURN REDUCTION

≈ **\$5M**  
ANNUAL SAVINGS

Higher retention,  
stronger customer  
lifetime value.